

Chap 11. Wavelets

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May 30, 2019

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Given a vector space of functions, one would like an orthonormal set of basis functions that span the space. For example, [Fourier transform](#):

$$f(x) = \int_0^{\infty} (A(w) \cos wx + B(w) \sin wx) dw$$

provides a set of basis functions based on sines and cosines.

However, for efficient computation, we would like the basis vectors to have **finite support** so that we could deal with local structure. Also we would like to have **easily computable coefficients** of the expansion of a function in the basis.

Now, we will follow the development of wavelets.

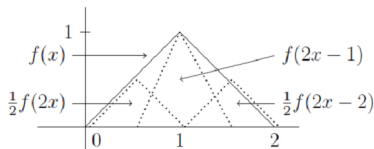
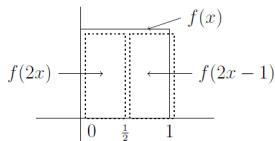
Dilation

A **dilation** is a mapping that scales all distances by the same factor. For example, a function f satisfying

$$f(x) = \sum_{k=0}^{d-1} c_k f(2x - k)$$

is a dilation. Some simple examples of dilation:

$$f(x) = \mathbf{1}_{[0,1)} \quad \text{or} \quad f(x) = \begin{cases} x & 0 \leq x < 1 \\ 2 - x & 1 \leq x < 2 \end{cases}$$



$$f(x) = f(2x) + f(2x - 1) \quad f(x) = \frac{1}{2}f(2x) + f(2x - 1) + \frac{1}{2}f(2x - 2).$$

Dilation

We say **all dilations in the equation are factor of two reductions** if the dilation equation has the form

$$f(x) = \sum_{k=0}^{d-1} c_k f(2x - k). \quad (1)$$

Lemma (11.1)

If a dilation equation (1) has a solution ϕ , then either $\sum_{k=0}^{d-1} c_k = 0$ or $\int_{-\infty}^{\infty} \phi(x) dx = 0$.

Proof.

By integrating both side of the dilation equation,

$$\int_{-\infty}^{\infty} \phi(x) dx = \sum_{k=0}^{d-1} \int_{-\infty}^{\infty} c_k \phi(2x - k) dx = \frac{1}{2} \sum_{k=0}^{d-1} c_k \int_{-\infty}^{\infty} \phi(x) dx.$$

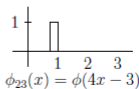
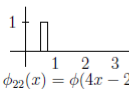
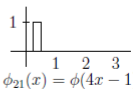
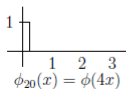
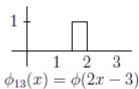
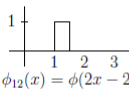
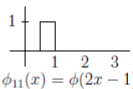
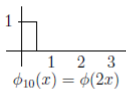
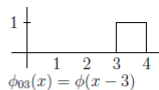
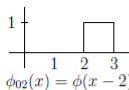
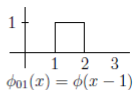
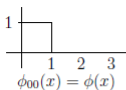


The Haar Wavelet

Let $\phi(x)$ be a solution to the dilation equation $f(x) = f(2x) + f(2x - 1)$.

The function ϕ is called a **scale function** or **scale vector** and is used to generate the two dimensional family of functions, $\phi_{jk}(x) = \phi(2^j x - k)$, where j and k are non-negative integers.

The **function space** V_j is defined with its basis $\{\phi_{jk}\}_{k \geq 0}$.



The Haar Wavelet

Some properties:

- ① $V_j \subseteq V_{j+1}$ since ϕ_{jk} is a linear combination of $\phi_{j+1,k}$ s.
- ② ϕ_j s are orthogonal, i.e. $\int \phi_{jk}(x)\phi_{jl}(x)dx = 0$ for $k \neq l$.

Think of a space $\text{span}\{\phi_{jk}\}_{j,k \geq 0}$. Since $\phi_{jk}, \phi_{j+1,2k}, \phi_{j+1,2k+1}$ are linearly dependent, $\{\phi_{jk}\}_{j,k \geq 0}$ is an over-complete basis and is not orthogonal.

$\Rightarrow$ Delete $\phi_{j+1,k}$ for odd k s : complete basis.

$\Rightarrow$ Replace $\phi_{j,2k}$ with $\psi_{j+1,2k}$: orthogonal. Here,

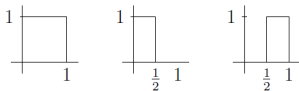
$$\psi_{jk}(x) = \begin{cases} 1 & \frac{2k}{2^j} \leq x < \frac{2k+1}{2^j} \\ -1 & \frac{2k+1}{2^j} \leq x < \frac{2k+2}{2^j} \end{cases}.$$

We call the two functions $\phi(x) = \phi_{00}(x)$ and $\psi(x) = \psi_{10}(x)$, the **Haar Wavelet**.

The Harr Wavelet

For instance,

Sub-basis of V_1



ϕ_{00}

ϕ_{10}

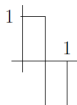
ϕ_{11}



ϕ_{00}



ψ_{10}



Sub-basis of V_3

ϕ_{00}

ϕ_{10}, ϕ_{11}

$\phi_{20}, \phi_{21}, \phi_{22}, \phi_{23}$

$\phi_{30}, \phi_{31}, \phi_{32}, \phi_{33}, \phi_{34}, \phi_{35}, \phi_{36}, \phi_{37}$



ϕ_{00}

ψ_{10}

ψ_{20}, ψ_{22}

$\psi_{30}, \psi_{32}, \psi_{34}, \psi_{36}$

The Harr Wavelet

Now, we can approximate a function with finite support by selecting a scale vector $\phi(x)$, and scale functions $\phi_{jk}(x) = \phi(2^j x - k)$, $k = 0, 1, \dots$, for some j . Dividing the x -axis into intervals of size 2^{-j} , approximate f as

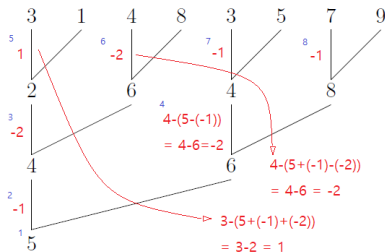
$$f(x) \approx \sum_{k=0}^{2^j-1} s_k \phi_{jk}(x) = \sum_{k=0}^{2^j-1} c_k \psi_{jk}(x).$$

Then, convert s_k s to c_k s, the coefficients for ψ_{jk} s. For instance,

$$\begin{pmatrix} 3 \\ 1 \\ 4 \\ 8 \\ 3 \\ 5 \\ 7 \\ 9 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & -1 & 0 & 0 & 0 \\ 1 & 1 & -1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & -1 & 0 & 0 & -1 & 0 & 0 \\ 1 & -1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & -1 & 0 & 1 & 0 & 0 & -1 & 0 \\ 1 & -1 & 0 & -1 & 0 & 0 & 0 & 1 \\ 1 & -1 & 0 & -1 & 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \\ c_6 \\ c_7 \\ c_8 \end{pmatrix}.$$

The Harr Wavelet

The last part can be easily done by using the **Tree of function averages**.



$$\Rightarrow (5, -1, -2, -2, +1, -2, -1, -1)$$

$$\begin{pmatrix} 3 \\ 1 \\ 4 \\ 8 \\ 3 \\ 5 \\ 7 \\ 9 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} - 1 \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ -1 \\ -1 \\ -1 \\ -1 \end{pmatrix} - 2 \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ -1 \\ 0 \\ 0 \\ 0 \end{pmatrix} - 2 \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \\ 0 \\ 0 \\ 1 \\ -1 \end{pmatrix} + 1 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} - 2 \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} - 1 \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} - 1 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ -1 \\ 0 \\ 0 \end{pmatrix} - 1 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ -1 \end{pmatrix}$$

Wavelet Systems

We now consider general **wavelet systems** with a basic **scaling function** $\phi(x)$ with a dilation equation

$$\phi(x) = \sum_{k=0}^{d-1} c_k \phi(2x - k),$$

and a **mother wavelet** $\psi(x)$. Assume $c_0 c_{d-1} \neq 0$. Scaling and shifting of the basic scaling function and the mother wavelet gives a set of **scaling functions** ϕ_{jk} s and a set of **wavelets** ψ_{jk} s,

$$\phi_{jk}(x) = \phi(2^j x - k), \quad \psi_{jk}(x) = \psi(2^j x - k).$$

Let $V_j = \text{span}\{\phi_{jk}\}_{k \geq 0}$. Then, $V_0 \subseteq V_1 \subseteq V_2 \subseteq \dots$.

Now, we want to find a proper scaling function $\phi(x)$ and a mother wavelet $\psi(x)$ so that the basis $\{\phi, \psi_{jk}\}$ is complete and orthogonal.

Solving the Dilation Equation

Consider solving the dilation equation

$$\phi(x) = \sum_{k=0}^{d-1} c_k \phi(2x - k)$$

to obtain the scale function for a wavelet system.

Perhaps the easiest way is to assume a solution and then calculate the scale function by **successive approximation**. Assume the initial guess of the values of ϕ on linearly spaced interval. Then, update the values of ϕ using the dilation equation until it converges.

Another approach to solving the dilation equation is to iteratively **compute** $\phi(\frac{j}{2^j})$ for large j , until $\phi(x)$ is approximated to a desired accuracy.

Solving the Dilation Equation

ex) For the dilation equation $\phi(x) = \frac{1}{2}\phi(2x) + \phi(2x-1) + \frac{1}{2}\phi(2x-2)$, assume $\text{supp}(\phi) \subseteq [0, 2)$. Then,

$$\phi(0) = 0, \quad \phi(2) = 0, \quad \phi(1) = \phi(1) = \text{arbitrary}.$$

Set $\phi(1) = 1$. Then, we get

$$\phi\left(\frac{1}{2}\right) = \frac{1}{2}\phi(1) = \frac{1}{2}, \quad \phi\left(\frac{3}{2}\right) = \frac{1}{2}\phi(1) = \frac{1}{2},$$

and

$$\phi\left(\frac{1}{4}\right) = \frac{1}{2}\phi\left(\frac{1}{2}\right) = \frac{1}{4}, \quad \phi\left(\frac{3}{4}\right) = \frac{1}{2}\phi\left(\frac{3}{2}\right) + \phi\left(\frac{1}{2}\right) = \frac{3}{4}.$$

One can continue this process and compute $\phi\left(\frac{j}{2^i}\right)$ for larger values of j until $\phi(x)$ is approximated to a desired accuracy.

Conditions on the Dilation Equation

Next lemma is a necessary but not sufficient condition on the coefficients of the dilation equation for shifts of the scale function to be **orthogonal**.

Lemma (11.2 orthogonality of $\phi(x)$ and $\phi(x - k)$)

If $\int_{-\infty}^{\infty} \phi(x)\phi(x - k)dx = \delta(k)$ for all k , then $\sum_{i=0}^{d-1} c_i c_{i-2k} = 2\delta(k)$.

Proof.

$$\begin{aligned} 2\delta(k) &= 2 \int_{-\infty}^{\infty} \phi(x)\phi(x - k)dx = 2 \sum_{i,j} c_i c_j \int_{-\infty}^{\infty} \phi(2x - i)\phi(2x - 2k - j)dx \\ &= \sum_{i,j} c_i c_j \int_{-\infty}^{\infty} \phi(y)\phi(y - (2k + j - i))dy = \sum_{i,j} c_i c_j \delta(2k + j - i) \\ &= \sum_{i=0}^{d-1} c_i c_{i-2k}. \end{aligned}$$



Conditions on the Dilation Equation

Lemma (11.3 support of ϕ)

Assume $\phi(x) = \sum_{k=-\infty}^{\infty} c_k \phi(2x - k)$. If $\text{supp}(\phi) \subseteq [0, d)$ and the set $\{\phi(x - k) : k \geq 0\}$ is linearly independent, then $c_k = 0$ unless $0 \leq k \leq d - 1$.

Proof. Omitted.

From the lemmas above, we can conclude that d is even.

Now we have $\frac{d}{2} + 1$ equations for d coefficients $c_0, c_1, \dots, c_{d-1}$,

$$\sum_{i=0}^{d-1} c_i = 0, \quad \sum_{i=0}^{d-1} c_i c_{i-2k} = 2\delta(k) \quad \text{for } 0 \leq k \leq \frac{d}{2} - 1.$$

Remark that these are not sufficient conditions for an appropriate scale function.

Derivation of the Wavelets from the Scaling Function

Now we want to derive the mother wavelet $\psi(x)$ as a linear combination of integer shifts of a scaled version of the scale function $\phi(x)$,

$$\psi(x) = \sum_{k=0}^{d-1} b_k \phi(2x - k).$$

Lemma (11.4 orthogonality of $\psi(x)$ and $\psi(x - k)$)

Assume the orthogonality of $\phi(x)$ and $\phi(x - k)$ for all k . If $\int_{-\infty}^{\infty} \psi(x)\psi(x - k)dx = \delta(k)$ for all k , then $\sum_{i=0}^{d-1} b_i b_{i-2k} = 2\delta(k)$.

Proof.

Analogous to Lemma 11.2. □

Derivation of the Wavelets from the Scaling Function

Lemma (11.5 orthogonality of $\phi(x)$ and $\psi(x - k)$)

Assume the orthogonality of $\phi(x)$ and $\phi(x - k)$ for all k . If $\int_{-\infty}^{\infty} \phi(x)\psi(x - k)dx = 0$ for all k , then $\sum_{i=0}^{d-1} c_i b_{i-2k} = 0$.

Proof.

$$\begin{aligned} 0 &= 2 \int_{-\infty}^{\infty} \phi(x)\psi(x - k)dx = 2 \sum_{i,j} c_i b_j \int_{-\infty}^{\infty} \phi(2x - i)\phi(2x - 2k - j)dx \\ &= \sum_{i,j} c_i b_j \int_{-\infty}^{\infty} \phi(y)\phi(y - (2k + j - i))dy = \sum_{i,j} c_i b_j \delta(2k + j - i) \\ &= \sum_{i=0}^{d-1} c_i b_{i-2k}. \end{aligned}$$



Derivation of the Wavelets from the Scaling Function

Lemma (11.6 overall)

Assume the orthogonality of $\phi(x)$ and $\phi(x - k)$, and of $\psi(x)$ and $\psi(x - k)$, and of $\phi(x)$ and $\psi(x - k)$ for all k . Then $b_k = (-1)^k c_{d-1-k}$ for all k . (or change all the signs, $-\psi(x)$)

Proof.

Consider the polynomials,

$$C_e(x) = c_0 + c_2x + \cdots + c_{d-2}x^{\frac{d}{2}-1}, \quad C_o(x) = c_1 + c_3x + \cdots + c_{d-1}x^{\frac{d}{2}-1},$$

$$B_e(x) = b_0 + b_2x + \cdots + b_{d-2}x^{\frac{d}{2}-1}, \quad B_o(x) = b_1 + b_3x + \cdots + b_{d-1}x^{\frac{d}{2}-1}.$$

Then, from the previous lemmas,

$$\begin{aligned} \sum_{j=0}^{d-1} c_j b_{j-2k} &= 0 & C_e(x)B_e(x^{-1}) + C_o(x)B_o(x^{-1}) &= 0 \\ \sum_{j=0}^{d-1} c_j c_{j-2k} &= 2\delta(k) & \rightarrow C_e(x)C_e(x^{-1}) + C_o(x)C_o(x^{-1}) &= 2 \\ \sum_{j=0}^{d-1} b_j b_{j-2k} &= 2\delta(k) & B_e(x)B_e(x^{-1}) + B_o(x)B_o(x^{-1}) &= 2 \end{aligned}$$

Derivation of the Wavelets from the Scaling Function

Proof (cont'd).

Thus, we get

$$\begin{pmatrix} C_e(x) & C_o(x) \\ B_e(x) & B_o(x) \end{pmatrix} \begin{pmatrix} C_e(x^{-1}) & B_e(x^{-1}) \\ C_o(x^{-1}) & B_o(x^{-1}) \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix},$$

and its determinants $D(x)D(x^{-1}) = 4$. Hence $D(x) = \pm 2x^m$ for some $m \geq 0$. WLOG, $D(x) = -2x^m$. Then,

$$\begin{aligned} 2B_o(x) &= C_e(x^{-1})D(x) = -2x^m C_e(x^{-1}), \\ 2C_o(x) &= -B_e(x^{-1})D(x) = 2x^m B_e(x^{-1}). \end{aligned}$$

Thus we get $m = \frac{d}{2} - 1$ and $b_{2k+1} = -c_{d-2k-2}$, $c_{2k} = b_{d-2k-1}$, which means that $b_k = (-1)^k c_{d-1-k}$.



Sufficient Conditions for the Wavelets to be Orthogonal

One would like a wavelet system to satisfy certain conditions:

- 1 All wavelets, $\psi_{jk} = \psi(2^j x - k)$, to be orthogonal to the scale function $\phi(x)$.
- 2 All wavelets to be orthogonal. i.e.,

$$\int_{-\infty}^{\infty} \psi_{jk} \psi_{lm} dx = \delta(j-l) \delta(k-m).$$

- 3 $\text{span}\{\phi(x), \psi_{jk} : j < l, \text{all } k\} = V_l = \text{span}\{\phi_{lk} : \text{all } k\}.$

Now we will see two lemmas which gives sufficient conditions on the wavelet coefficients b_k to satisfy the above conditions.

Remark) Actually all k is not all integers k . Consider only the functions with $\text{supp} \subseteq [0, d)$.

Remark) The third condition holds if the others hold. Orthogonal basis.

Sufficient Conditions for the Wavelets to be Orthogonal

Lemma (11.7 condition 1)

If $b_k = (-1)^k c_{d-1-k}$ for all k , then $\int_{-\infty}^{\infty} \phi(x) \psi(2^j x - l) dx = 0$ for all j, l .

Proof.

It is easy to show that $\phi(x)$ and $\psi(x - k)$ are orthogonal. Since $\int_{-\infty}^{\infty} \phi(2^j x - k) \psi(2^j x - l) dx = \frac{1}{2^j} \int_{-\infty}^{\infty} \phi(y - k) \psi(y - l) dy = 0$ and $\phi(x)$ is a linear combination of $\phi(2^j x - k)$ s, the lemma holds. □

Lemma (11.8 condition 2)

If $b_k = (-1)^k c_{d-1-k}$ for all k , then for all j and l ,

$$\int_{-\infty}^{\infty} 2^{j/2} \psi(2^j x - k) 2^{l/2} \psi(2^l x - m) dx = \delta(j - l) \delta(k - m).$$

Remark) One can define ψ_{jk} as normalized one as $2^{j/2} \psi(2^j x - k)$.

Sufficient Conditions for the Wavelets to be Orthogonal

Proof.

$$\begin{aligned}\int_{-\infty}^{\infty} \psi(x-k)\psi(x-m)dx &= \sum_{i,j} b_i b_j \int_{-\infty}^{\infty} \phi(2x-2k-i)\phi(2x-2m-j)dx \\ &= \frac{1}{2} \sum_{i=0}^{d-1} b_i b_{i-2k+2m} = \frac{1}{2} \sum_{j=0}^{d-1} c_j c_{j+2k-2m} = \delta(k-m).\end{aligned}$$

Let $l = j + t$ for some $t \geq 0$, then

$$\int_{-\infty}^{\infty} \psi(2^j x - k)\psi(2^l x - m)dx = \int_{-\infty}^{\infty} \frac{1}{2^j} \psi(y - k)\psi(2^t y - m)dy.$$

If $t = 0$, $RHS = \frac{1}{2^j} \delta(k - m)$.

If $t > 0$, $\psi(y - k)$ is a linear combination of $\phi(2y - s)$ s. Thus RHS is a linear combination of

$$\int_{-\infty}^{\infty} \phi(z)\psi(2^{t-1}z - r)dz = 0.$$

Thus, $RHS = 0$ and the lemma holds. □

Expressing a Function in Terms of Wavelets

Given wavelet system, we want to express a function $f(x)$ in terms of an orthonormal basis of the system. We first express $f(x)$ in terms of scale functions ϕ_{jk} , and then convert it in terms of wavelets ψ_{jk} by the **tree diagram** similar to the previous one.

Let $f(x) = \sum_{k=0}^{\infty} a_{jk} \phi_j(x - k) dx$. Then,

$$\begin{aligned} a_{jk} &= \int_{-\infty}^{\infty} f(x) \phi_j(x - k) dx \\ &= \sum_{m=0}^{d-1} \int_{-\infty}^{\infty} c_m f(x) \phi_{j+1}(2x - 2k - m) dx \\ &= \sum_{n=2k}^{d-1} c_{n-2k} a_{j+1,n}. \end{aligned}$$

Designing a Wavelet System

In designing a wavelet system there are a number of parameters in the dilation equation. As we saw before, c_i s need to satisfy $\frac{d}{2} + 1$ equations:

$$\sum_{i=0}^{d-1} c_i = 2,$$

and

$$\sum_{i=0}^{d-1} c_i c_{i-2k} = \delta(k).$$

The remaining $\frac{d}{2} - 1$ degrees of freedom can be used to obtain some desirable properties such as smoothness.

Wavelets are widely used in **data compression** for images and speech, as well as in computer vision for **representing images**. Unlike the sines and cosines of the Fourier transform, wavelets have spatial locality in addition to frequency information, which can be useful for better understanding the contents of an image and for relating pieces of different images to each other. Wavelets are also being used in **power line communication protocols** that send data over highly noisy channels.

The End